

Thaaher

Machine Learning Engineer | MLOps, Generative AI & Production ML Systems
d@gmail.com

SUMMARY

Senior Data Scientist and Machine Learning Engineer specializing in production-grade ML systems, Generative AI, LLMs, RAG, and MLOps. Experienced in designing, deploying, and maintaining scalable AI/ML pipelines, API-based model services, cloud-native workflows, and enterprise AI products across AWS, Azure, and GCP. Skilled in collaborating with data scientists and engineering teams to productionize models using FastAPI, Flask, Docker, CI/CD, MLflow, vector databases, and orchestration frameworks. Proven ability to transform experimental models into secure, reliable, maintainable AI-enabled software solutions with strong focus on automation, monitoring, deployment efficiency, and business impact.

SKILLS

Machine Learning, Deep Learning, MLOps, LLMops, Generative AI, Large Language Models, Natural Language Processing, Retrieval-Augmented Generation, Model Deployment, Production ML Pipelines, ML Workflow Automation, Model Lifecycle Management, Model Monitoring, Data Drift Detection, Explainable AI, Predictive Modeling, Time Series Analysis, Recommendation Systems, Computer Vision, Python, Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, OpenCV, NLTK, Pandas, NumPy, PySpark, FastAPI, Flask, Streamlit, REST APIs, GraphQL, Docker, GitHub Actions, CI/CD Pipelines, MLflow, ZenML, LangChain, LangGraph, LlamaIndex, LangSmith, AutoGen, AutoML, OpenAI API, AWS SageMaker, AWS Lambda, AWS Bedrock, AWS EC2, Google Vertex AI, Google Cloud Functions, Google Cloud Run, BigQuery, Azure ML, Azure AI, Azure App Services, SQL, NoSQL, Pinecone, Faiss, Chroma DB, Neo4j, Agile Methodologies, Microservices Architecture, Stakeholder Collaboration, Team Leadership

EXPERIENCE

Senior Data Scientist & Machine Learning Software Engineer, Generative AI - PURELOGICS

12/2024 – Present

- Designed, deployed, and optimized end-to-end AI/ML pipelines on AWS SageMaker, Bedrock, Lambda, GCP Vertex AI, BigQuery, and Cloud Run.
- Built enterprise-grade RAG pipelines, conversational AI systems, health monitoring AI workflows, and agentic AI solutions for production environments.
- Collaborated with cross-functional teams to align AI architecture, model lifecycle management, and deployment strategies with business objectives.
- Implemented scalable LLMops and MLOps workflows to support development, deployment, automation, and continuous improvement of AI products.
- Engineered multi-agent AI architectures using LangChain, LangGraph, PhiData, Pinecone, and graph databases to automate decision-making workflows.
- Mentored AI/ML teams on LLM fine-tuning, Retrieval-Augmented Generation, and production-ready AI automation practices.

Senior Data Scientist, Machine Learning & Generative AI Engineer - NVIDIA

08/2024 – Present

- Contributed to GPU-optimized AI applications and enterprise Generative AI solution testing through NVIDIA and Z by HP developer programs.
- Developed and tested open-source and enterprise-level LLMs, including vLLM-based solutions for Generative AI use cases.

- Integrated NVIDIA tools and SDKs to build and optimize AI solutions for large-scale deployment scenarios.
- Collaborated with NVIDIA and Z by HP teams to validate AI tools, frameworks, and performance improvements.
- Supported open-source LLM initiatives focused on Generative AI innovation and deployment readiness.

Senior Data Scientist & Machine Learning Engineer - BLOCBELT

12/2022 – 12/2024

- Led development of AI-driven SaaS and PaaS products, managing a team of three across data science, AI engineering, and big data initiatives.
- Delivered AI solutions that improved operational efficiency and generated \$2M in cost savings for clients.
- Improved sales forecasting accuracy by 80–90% through advanced data analysis across 120+ market entities.
- Launched a personalized fashion e-commerce platform with a recommendation engine, increasing sales by 80% and online orders by 200%.
- Developed statistical models and predictive analytics solutions that contributed to an 18% revenue increase.
- Enhanced data visibility by 45% through strategic dashboards and cross-functional analytics enablement.
- Integrated AI capabilities into SaaS and PaaS offerings, using distributed systems and 25+ A/B tests to optimize digital sales experiences.

AI Data Scientist Kaggle Master Mentor - GOOGLE-KAGGLE

12/2022 – 11/2024

- Mentored aspiring data scientists through the Kaggle-X BIPOC Mentorship Program, dedicating 20 hours weekly to technical development.
- Ranked among the Top 100 Kaggle contributors globally and recognized as the first Pakistani mentor in the program.
- Led a team of four mentees in machine learning, data science, and real-world problem-solving practices.
- Collaborated on high-impact AI projects and contributed to the global data science community.

Data Scientist - Crypto-Express

01/2022 – 12/2022

- Developed an anti-spoofing face application to strengthen digital identity security and reduce identity fraud by 70%.
- Applied machine learning and deep learning techniques to improve operational efficiency by 25% and reduce costs by 15%.
- Built predictive models and delivered insights that supported a 20% revenue increase through data-driven decision-making.
- Automated exploratory data analysis and machine learning workflows, saving 250+ hours of manual analysis annually.
- Deployed AI/ML projects remotely while adapting solutions to changing business and technical requirements.

Artificial Intelligence Engineer - Pakistan Freelancing Training Center

01/2021 – 12/2021

- Led AI and data science training sessions focused on practical machine learning applications.
- Designed and deployed TensorFlow and Keras models to improve system efficiency and accuracy.
- Automated repetitive data tasks using Python to reduce manual effort and error rates.
- Supported applied AI research and translated emerging methodologies into practical project work.

PROJECTS

AutoML-Studio

Built a low-code/no-code machine learning platform for rapid predictive model development and deployment. Integrated MLflow for experiment tracking, SHAP for explainability, and data drift detection for model reliability. Deployed with FastAPI and Streamlit to support classification, regression, time-series forecasting, and clustering workflows.

Customer Satisfaction Prediction System

Created a predictive analytics system using ZenML and MLflow within an MLOps framework, achieving 93% accuracy in forecasting customer satisfaction. Streamlined pipeline management and model deployment to support actionable business decision-making and customer retention strategies.

Conversational AI Chatbot for Customer Support

Designed and deployed an enterprise-grade chatbot powered by GPT-4o, LangChain, and AWS Bedrock. Implemented intent recognition and dynamic response generation, reducing customer query resolution time by 40% and improving customer satisfaction.

DocuWiz AI

Engineered a document intelligence tool using BART, DistilBERT, RAG, and LLMs for text extraction, summarization, sentiment analysis, and document insights. Deployed with Streamlit and FastAPI, reducing document processing time by 60% for legal and academic workflows.

Health Assistance Application

Architected a healthcare AI application using Gemini 1.5 Pro, RAG, and LangChain to support doctors and patients with medical information, symptom analysis, treatment suggestions, and preventive healthcare guidance.

AI-Powered Blog Generator

Developed an AI content generation system using Llama 3.1 8B, AWS Bedrock, and RAG for dynamic dataset-based content creation. Integrated Streamlit to automate blog creation, editing, and publishing workflows.

Harvestify AI-Powered Plant Health Assistant

Designed and deployed an Azure-based computer vision application for real-time plant leaf disease diagnosis, crop recommendations, and fertilizer suggestions. Reduced disease detection time by 70% and improved crop yield predictions by 30%.

Intelligent Document Summarization Tool

Developed a transformer-based summarization system using Pegasus to generate concise summaries from PDF and DOCX documents. Deployed with FastAPI and Streamlit, reducing manual document review time by 60%.

EDUCATION

Bachelor in Computer Science, Major: Artificial Intelligence and Computer Science - Institute of Management Sciences